Miracle Apata — Desktop Software Engineer

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Summary

Experienced desktop software engineer specializing in cross-platform applications for **Windows** and **macOS** using **Python (PySide6)** and **SwiftUI**. Adept at building multilingual educational tools used by universities, academies, and institutions for standardized testing. Expert in packaging with PyInstaller and Inno Setup and delivering native desktop experiences.

Work Experience

DINATEN — Santa Cruz de Tenerife, Spain
https://dinaten.com
Aug 2023 – Present

- Developed and maintained piXeval and **pixEBAP**, two educational desktop applications for managing and correcting multiple-choice exams.
- Built apps using **PySide6** (Qt for Python), compiled with **PyInstaller**, and packaged installers with **Inno Setup**.
- Created and maintained a **macOS-native version of piXeval** using **SwiftUI**, designed specifically for Apple users while ensuring feature parity with the Windows build.
- Ensured full cross-platform support across **Windows and macOS** with consistent UX.
- Supported multiple languages in the app interface: **Catalan**, **Spanish**, **English**, **Euskera**, and **French**.
- Integrated key features:
  - Bulk import from Word, Excel, and Moodle
  - Generation of up to 9 randomized models of the same exam
  - Scanning and automatic grading of answer sheets (up to 30 exams/minute) using OCR and OMR techniques implemented with OpenCV2, TensorFlow, and other supporting libraries
  - Delivery of student results via email
  - Advanced statistics and competency evaluation

Skills

- **Languages:** Python, Swift (SwiftUI), SQL
- **Frameworks & Libraries:** PySide6, SwiftUI, OpenCV2, TensorFlow
- **Tools & Packaging:** PyInstaller, Inno Setup, Git, macOS development tools
- **Platforms:** Windows, macOS
- **App Languages:** Spanish, English, Catalan, Euskera, French
- **Tech Focus:** OCR/OMR, automated exam grading, cross-platform application development

Projects

- **piXeval**
  - Download
  - Intelligent exam correction and evaluation tool for teachers and institutions.
- **pixEBAP**
  - Internal desktop application used for managing exam forms and student data.

Personal Projects

- **SnapGrade** *(solo project, in active development)*
  - Desktop app for generating, scanning, and grading multiple-choice exam sheets via OMR — no Scantron hardware, no proprietary format lock-in.
  - Built the sheet-generation pipeline: auto-computed bubble-grid layouts (never too small to fill, never overflowing the page), barcode/QR-encoded sheet identity, and print-ready PDF export for single sheets or full batches.
  - Built and calibrated the OMR grading engine with **OpenCV**: barcode/QR decoding resolves each sheet's identity and expected layout, bubbles are classified by measured ink-fill ratio rather than fixed guesses, and ambiguous answers are flagged for review instead of silently guessed.
  - Implemented Excel-based roster import (prefilled sheets, ID-matched grading) and results export (summary + per-question detail) with **openpyxl**.
  - Built the **PySide6** UI with a token-based theming system — five switchable themes generated from one shared QSS template.
  - Sheet generation, the grading engine, and Excel I/O are built and validated against a calibration fixture; wiring the UI to those engines is the one remaining piece.

  Tech: PySide6, OpenCV, pyzbar, qrcode, python-barcode, ReportLab, openpyxl, uv

Education

B.Eng. in Electrical & Electronics Engineering
Federal University of Technology, Akure — Graduated 2024

Languages

- English (Fluent, C1)
- Spanish (Conversational, B1)

Contact

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